

SREE GNANESH DOMMETI

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Education

University of Maryland, College Park

M.S. in Applied Machine Learning; GPA: 3.9 / 4.0

May 2026

College Park, MD

National Institute of Technology, Calicut

B.Tech. in Computer Science and Engineering; GPA: 8.21 / 10

May 2022

Calicut, India

Experience

GAMMA Lab, University of Maryland

January 2026 – May 2026

Research Assistant, Advisor: Dinesh Manocha

College Park, MD

- Contributed to ongoing work on grounded multimodal/video reasoning evaluation, decomposing model outputs into reasoning steps and checking each step against visual/textual evidence.
- Developed trace-level checks to expose unsupported intermediate claims, hallucinated visual references, and weak evidence chains in multimodal responses.

FROOT Lab, University of Maryland

November 2025 – Present

Research Assistant, Advisor: Alan Zaoring Liu

College Park, MD

- Contributing to ProjectASAP, an approximation-first telemetry research effort that accelerates SQL/PromQL analytics for agent-era data pipelines.
- Built PromQL workload-analysis, sketch-planning, and streaming-summary components across Prometheus, Arroyo, Kafka, and Rust query-serving paths.
- Built NetArena benchmark tooling for the MALT capacity-planning task with Dockerized LiteLLM prompts, output validation, retries, and reproducible local submissions.

Qualcomm

July 2022 – July 2024

Software Engineer

Hyderabad, India

- Owned LTE Layer 3 RRC Cell Selection and Security modules for 4G/5G, CAT-M1, and NB-IoT modem stacks.
- Built Python log-analysis and regression-triage tooling for changelist assignment, RRC ownership routing, and diagnostic dashboards, reducing manual engineering effort by 40%.

Projects

TCOVITA - Temporally Consistent Video Instance Segmentation | *PyTorch, Mask2Former, VITA, InfoNCE*

- Extended VITA's object-token paradigm with a Temporal Association Module using Hungarian matching over cosine similarity and IoU, plus InfoNCE contrastive loss for stable cross-frame identity assignment.
- Implemented training and evaluation on a shared HPC cluster; benchmarked on YouTube-VOS 2019 with 3,471 train / 507 validation videos and 65 object categories.
- Reduced ID switches by 32.8%, improved Association Accuracy by 41.4%, and decreased track fragmentation by 42.1% versus VITA while maintaining segmentation quality (J=0.63, F=0.61).

DiffSplat: 3D Gaussian Generation and Editing | *PyTorch, CUDA Rasterization, 3DGS, SAM, Inpaint360GS*

- Set up DiffSplat text-to-3D Gaussian generation with released checkpoints, compiled the CUDA rasterizer, and patched renderer / multi-view attention compatibility in A100 environments.
- Ran CLIP-based ablations across SD 1.5, PixArt-Sigma, SD 3.5 M, guidance scale, seed sensitivity, prompt category, and native 3D versus independent 2D views.
- Built a SAM 3 + Inpaint360GS object-removal extension with Gaussian pruning, virtual-view inpainting, and depth repair; reached masked PSNR 32.77, SSIM 0.9855, LPIPS 0.0082 at 12K iterations.

DistTrain Lab - LLM Pretraining & Post-Training Lab | *Python, TRL/Accelerate, FSDP, DeepSpeed*

- Added causal-LM pretraining plans for an 8x L40 target with data-curation manifests, LR/batch/precision sweeps, validation-loss/perplexity gates, checkpoint checks, and deterministic smoke runs.
- Implemented SFT/DPO/PPO/GRPO-style post-training artifacts, memory/throughput estimators, parallelism sweeps, quality gates, and DeepSpeed/FSDP/Megatron/TorchTitan-style configs.

Skills

ML / Research: PyTorch, TensorFlow, JAX, Scikit-learn, HuggingFace Transformers, Mask2Former, VITA, 3D Gaussian Splatting, SAM, CLIP, causal-LM pretraining workflows, LoRA, DPO, PPO/GRPO-style rewards, RAGAS

Distributed Training: TRL, Accelerate, Ray, DeepSpeed, FSDP, Megatron/TorchTitan configs, data/tensor/pipeline parallelism, activation checkpointing, 8-GPU L40 launch configs

Data / Systems: Python, SQL, C, C++, Rust, Kafka, Arroyo, Prometheus, Grafana, Elasticsearch, FAISS, Pinecone, PostgreSQL

MLOps / Infra: Docker, Kubernetes, AWS EC2/S3/SageMaker/ECS, MLflow, DVC, ClearML, W&B, GitHub Actions, OpenTelemetry